

Data Analytics (DA)

Module -3

MySQL Assignment 1 -

DDL Commands, Constraints

Assignment Task Overview

Problem Statement:

As a Data Analyst, the company requires an **Employee Database** to store and manage information related to employees, departments, and locations within the organization.

You are required to design and manage this employee database using **MySQL DDL commands**, focusing on creating, modifying, and managing database objects while enforcing appropriate constraints to ensure data integrity and consistency.

Dataset Description:

You will work to create the following three tables:

- **Table: *Departments***

Attributes:

- *department_id* – Primary Key
- *department_name*

- **Table: *Location***

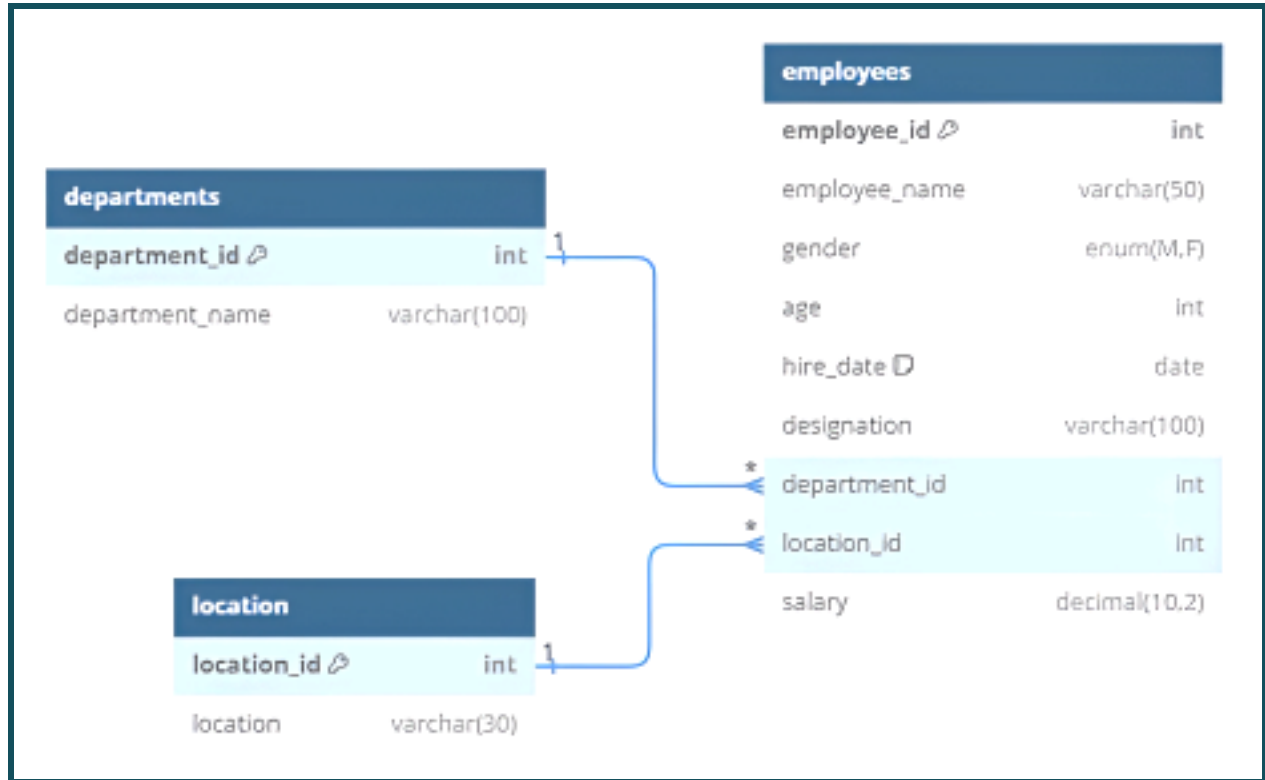
Attributes:

- *location_id* – Primary Key
- *location_name*

- **Table: *Employees***

Attributes:

- *employee_id* – Primary Key
- *Employee_name*
- *Gender*
- *Age*
- *Hire_date*
- *Designation*
- *Salary*
- *department_id* – Foreign Key referencing department(*department_id*)
- *location_id*—Foreign Key referencing location(*location_id*)



Tasks:

DDL Commands

1. Database and Table Creation (CREATE):

- Create a Database for this project
- Create all three tables in MySQL with appropriate **data types** and **relationships**.

2. Table Alteration (ALTER)

- Add a new column named "email" to the Employees table to store employee email addresses.
- Modify the data type of the "designation" column in the Employees table to support a wider range of values.
- Drop the "age" column from the Employees table.
- Rename the "hire_date" column to "date_of_joining".

3. Table Renaming (RENAME):

- Rename the "Departments" table to "Departments_Info".
- Rename the "Location" table to "Locations".

4. Table Truncation (TRUNCATE):

- Truncate the Employees table.

5. Database & Table Dropping (DROP):

- Drop the Employees table and then the "employee" database.

Tasks: Constraints -

1. Database Recreation:

- Drop the 'employee' database if it exists
- Recreate it using the provided schema, ensuring that all tables are created with the appropriate constraints as instructed.
- Establish links between the "department_id" and "location_id" fields in the "employees" table and their respective tables.

2. Departments Table:

- Ensure that the "department_id" **uniquely** identifies each department.
- Set up constraints on the "department_name" to avoid **duplicate and null entries**.

3. Locations Table:

- Establish a mechanism to automatically generate **unique identifiers** for each location, ensuring that they are **incremented sequentially**.
- Implement constraints to prevent the insertion of **null and duplicate locations**.

4. Employees Table:

- Guarantee that each employee has a distinct **identifier**.
- Create a restriction to ensure that the employee's name is always **provided**.
- Limit the acceptable values for the "**gender**" field to only 'M' or 'F'.
- Enforce a condition to ensure that the employee's **age is 18 or above**.
- Automatically assign the **current date to the "hire_date"** field if not specified.

Deliverables

Submit a project folder containing:

1. **MySQL script (.sql)** that includes:

- DDL commands for Table creation statements
- Tables are created with the appropriate constraints

2. **PDF Document** that includes:

- Execution screenshot

Skills to be Evaluated and Mapping

Assignment Section	Core Skill	Sub-Skill	Proficiency
Database Creation	Database Setup	SQL Environment Management	Beginner
Table Relationships	Relational Database Design	Primary & Foreign Keys	Beginner
Schema Modification	Database Maintenance	ALTER Commands	Beginner
Table Management	Data Definition	RENAME & TRUNCATE	Beginner
Database Cleanup	Database Administration	DROP Commands	Beginner
Data Integrity	Constraints	NOT NULL, UNIQUE, CHECK, DEFAULT	Intermediate
Auto Increment Logic	Key Management	AUTO_INCREMENT	Beginner
Validation Rules	Data Quality	Conditional Constraints	Intermediate
Referential Integrity	Data Relationships	Foreign Key Constraints	Intermediate

DA Skill Taxonomy Structure

Skill / Component	Full Marks	Criteria for Full Marks	Partial Credit Description	Partial Marks Range
Database Creation, Table Creation & Relationships	3	Database created successfully using CREATE DATABASE All three tables created with correct data types and foreign key relationships	Database created with minor syntax issues Tables created but relationships or data types incorrect	1-2
Table Alteration (ALTER)	2	All ALTER operations executed correctly (add, modify, drop, rename columns)	One or more ALTER operations incorrect or missing	1
Table Management (RENAME, TRUNCATE, DROP)	2	Tables renamed, truncated, and dropped correctly	Minor mistakes in table management commands	1
Constraints Implementation	3	All required constraints applied correctly across all tables	Some constraints missing or incorrectly applied	1 – 2

DA Skill Taxonomy Structure

- **Database Setup:** Create MySQL database, tables, and define relationships
- **DDL Operations:** Use CREATE, ALTER, RENAME, TRUNCATE, and DROP commands
- **Table Modification:** Add, modify, delete, and rename table columns
- **Constraint Enforcement:** Apply PRIMARY KEY, FOREIGN KEY, UNIQUE, NOT NULL, CHECK, and DEFAULT constraints

Evaluation Guidelines

- **Open and review the submitted .sql script** to ensure:
 - Tables are correctly created and populated.
 - All data exploration Using Joins and five analytical queries are present and executable.
- **Check the correctness of joins, grouping, and calculations.**
- **For each section**, match student work to the specific skill/sub-skill area in the taxonomy.
- **Assign full or partial marks** according to the rubric.
 - Full marks: accurate syntax, logic, and clean formatting.
 - Partial marks: minor issues, incomplete queries, or inconsistent naming.
- **Comment** on strengths and improvement points for each section.
- **Maintain objectivity**—assess correctness, clarity, and relevance of SQL queries.

Achievement Criteria & Passing Rules

- At least partial (non-zero) marks must be obtained in every assignment section.
- Students must fully achieve at least two major skills: DDL Commands & Constraints
- All required SQL scripts must be executable and logically correct.
- Feedback and a detailed scorecard must be provided for every evaluated submission.